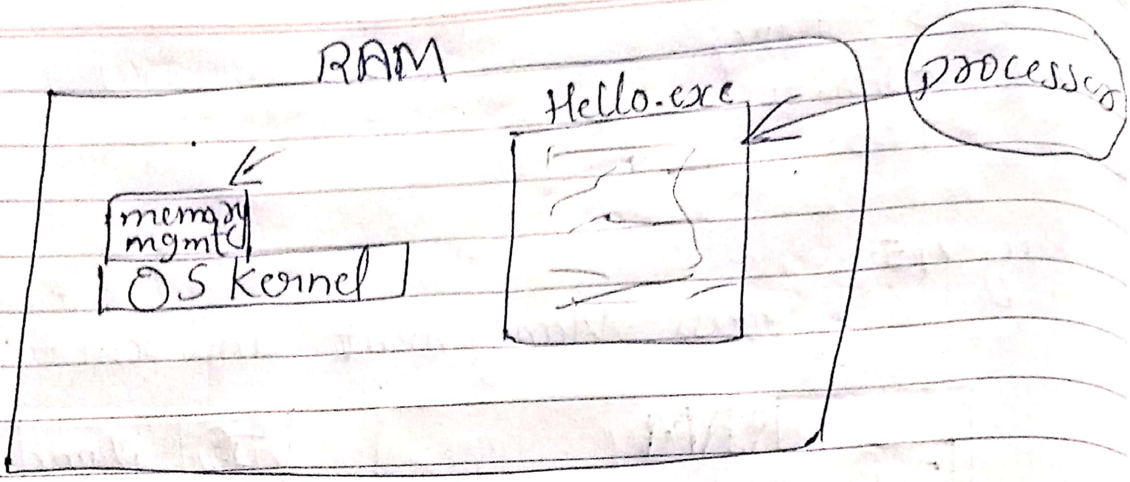
Day 46: Memory Management

- Allocation / deallocation of memory to processes
- Safe and efficient memory usage
- Isolation and protection between processes

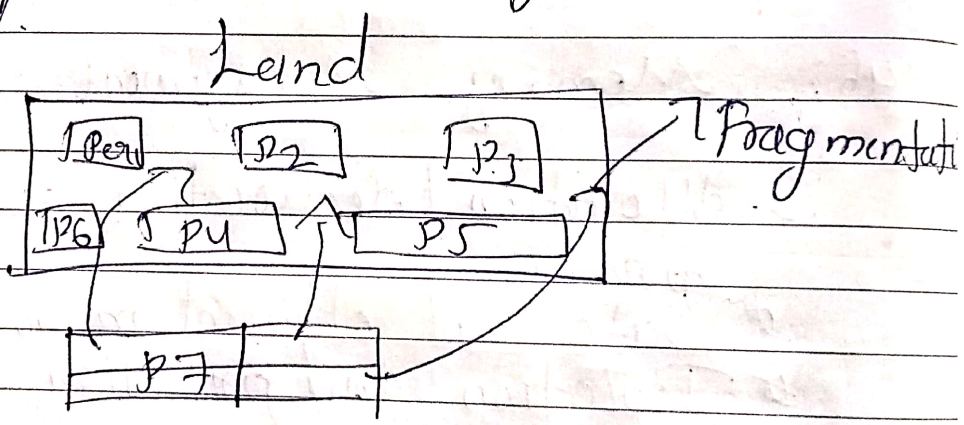


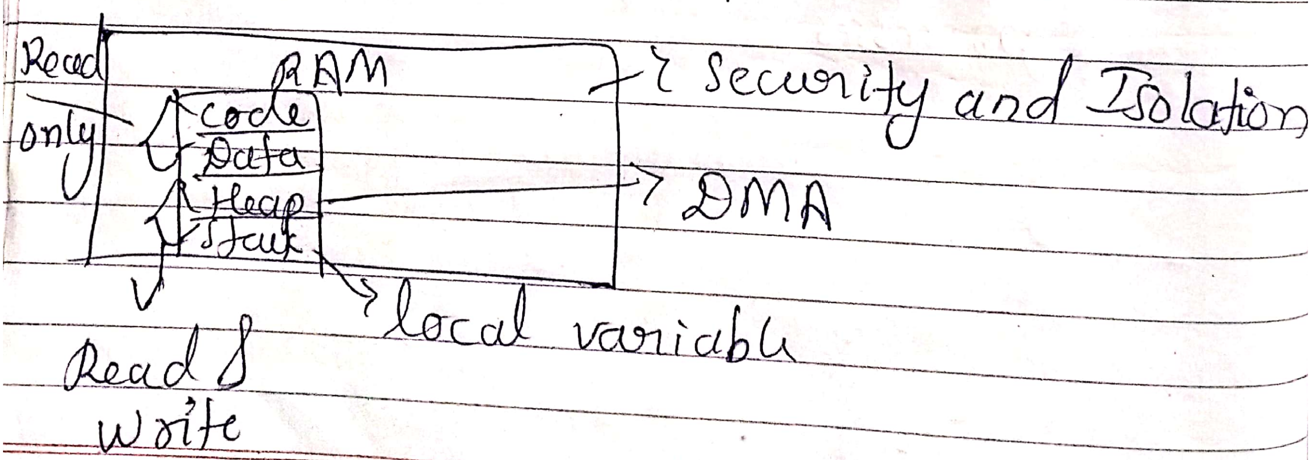
1) Segmentation → 32bit old, Embedded
2) Paging → Modern, 64 bit, Linux, win, mac

these are memory allocation algorithms

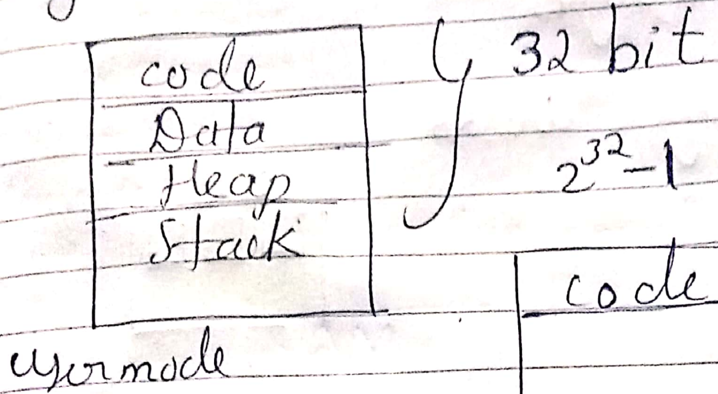
Why?

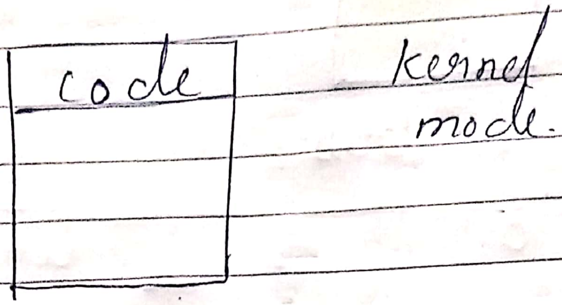
Imagine Land seller, sell his land to customer by the parts of land.





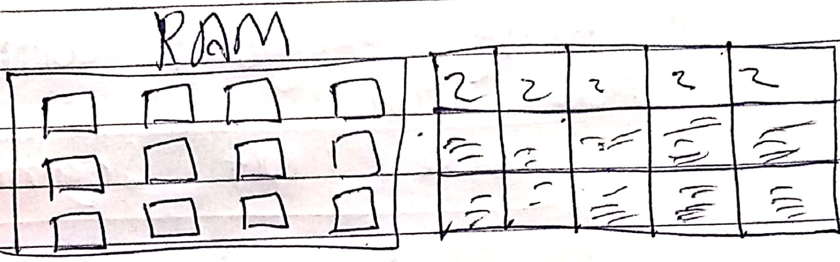
Segmentation:





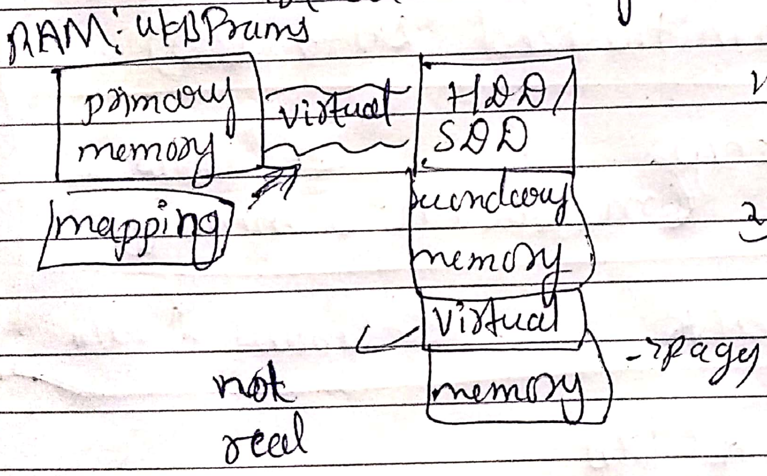
Paging Drawback of segmentation
1) fragmentation,
2) no more than 4GB memory.

Paging:



Virtual Memory why:

RAM: 4GB rams



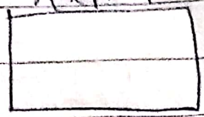
1) RAM is very expensive
2) Software is very large

~~Why~~ virtual memory gives space in pages -> each pages has fixed size.
4KB

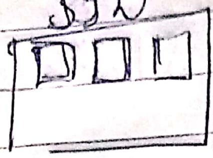
process Table
Frames -> pages

How paging works

RAM



SSD



address

0 1 2 3 4 5 6

base=0 base=2 -> offset

page fault

Memory Layout with Paging

Low address

code/Text Segment
Initialized Data Segment
Uninitialized Data (BSS)
Heap <= Grows up (malloc/new memory)
Stack <= Grows Down

High Address

Paging Algorithms

→ FIFO — Replace oldest page

→ LRU — Replace least recently accessed page

→ optimal → Replace ~~algorith~~ page that won't be used for longest time (ideal)

clock → circular queue with reference bits (2nd chance)